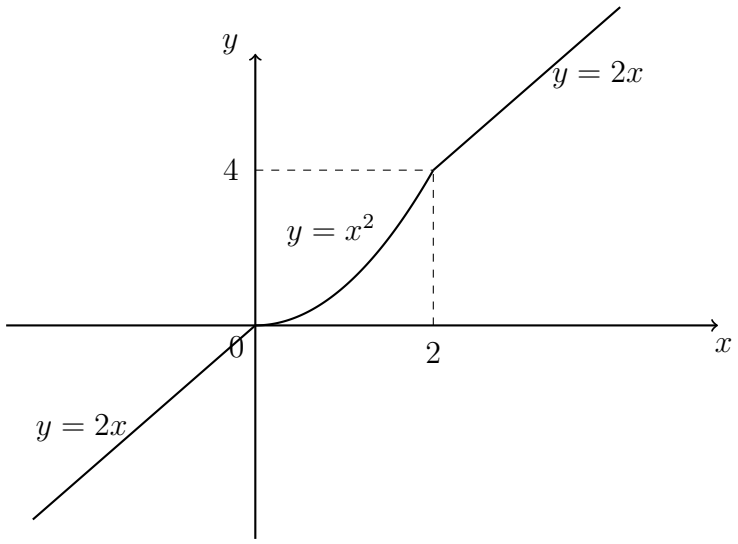
(i)

$$\text{Min}(x^2, 2x) = \begin{cases} x^2, & x^2 \leq 2x \\ 2x, & x^2 > 2x \end{cases}$$

$$x^2 \leq 2x \implies x(x-2) \leq 0 \implies 0 \leq x \leq 2$$
$$x^2 > 2x \implies x > 2 \text{ or } x < 0$$

$$\text{Min}(x^2, 2x) = \begin{cases} x^2, & 0 \leq x \leq 2 \\ 2x, & x < 0 \text{ or } x > 2 \end{cases}$$



(ii)

$$2\text{Min}(x^2, 2x) = 5x - 3$$

$$\text{If } 0 \leq x \leq 2 \implies 2x^2 = 5x - 3$$
$$(2x-3)(x-1) = 0 \implies x = \frac{3}{2},\ 1$$

$$\text{If } x < 0 \text{ or } x > 2 \implies 4x = 5x - 3 \implies x = 3$$

Hence, $\boxed{x = 1,\ \frac{3}{2},\ 3}$

(iii)

$$\text{Min}(x^3, 4x) = \begin{cases} x^3, & x^3 \leq 4x \\ 4x, & x^3 > 4x \end{cases}$$

$$x^3 \leq 4x \implies x(x-2)(x+2) \leq 0$$

$$x \leq -2 \quad \text{or} \quad 0 \leq x \leq 2$$

$$\text{Min}(x^3, 4x) = \begin{cases} x^3, & x \leq -2 \text{ or } 0 \leq x \leq 2 \\ 4x, & -2 < x < 0 \text{ or } x > 2 \end{cases}$$

$$\text{Min}(x^2, 2x) = \begin{cases} x^2, & 0 \leq x \leq 2 \\ 2x, & x < 0 \text{ or } x > 2 \end{cases}$$

When $m = 2$

1. $x \leq -2: \quad x^3 + 2x = 2x$ (rejected)
2. $-2 < x < 0: \quad 4x + 2x = 2x$ (rejected)
3. $0 \leq x \leq 2: \quad x^3 + x^2 = 2x \implies x(x+2)(x-1) = 0$
4. $x > 2: \quad 4x + 2x = 2x$ (rejected)

So for $m = 2$: $\boxed{x = 0,\ 1}$

When $m = 6$

1. $x \leq -2: \quad x^3 + 2x = 6x \implies x(x-2)(x+2) = 0 \implies x = -2$
2. $-2 < x < 0: \quad 4x + 2x = 6x \implies x \in \mathbb{R}\ (-2 < x < 0)$
3. $0 \leq x \leq 2: \quad x^3 + x^2 = 6x \implies x(x+3)(x-2) = 0 \implies x = 0, 2$
4. $x > 2: \quad 4x + 2x = 6x \implies x \in \mathbb{R}$

So for $m = 6$: $\boxed{-2 \leq x \leq 0 \text{ or } x \geq 2}$

(iv)

$$y = 2\text{Min}(x^2, x^3) - 5x$$

$$y = \begin{cases} 2x^2 - 5x, & \text{when } x \geq 1 \\ 2x^3 - 5x, & \text{when } x < 1 \end{cases}$$

For $x \geq 1$

$$\frac{dy}{dx} = 4x - 5 = 0 \implies x = \frac{5}{4}$$

$$y = 2x^2 - 5x = 2 \cdot \frac{25}{16} - 5 \cdot \frac{5}{4} = -\frac{25}{8}$$

$$\frac{d^2y}{dx^2} = 4 > 0 \implies \left(\frac{5}{4}, -\frac{25}{8}\right) \text{ is minimum point}$$

For $x < 1$

$$\frac{dy}{dx} = 6x^2 - 5 = 0 \implies x = \pm\sqrt{\frac{5}{6}}$$

$$y = \frac{2}{3}\left(\pm\sqrt{\frac{5}{6}}\right)^3 - 5\left(\pm\sqrt{\frac{5}{6}}\right) = \mp\frac{10}{3}\sqrt{\frac{5}{6}}$$

$$\frac{d^2y}{dx^2} = 12x \implies \left(\sqrt{\frac{5}{6}}, -\frac{10}{3}\sqrt{\frac{5}{6}}\right) \text{ is minimum point}$$

$$\implies \left(-\sqrt{\frac{5}{6}}, \frac{10}{3}\sqrt{\frac{5}{6}}\right) \text{ is maximum point}$$

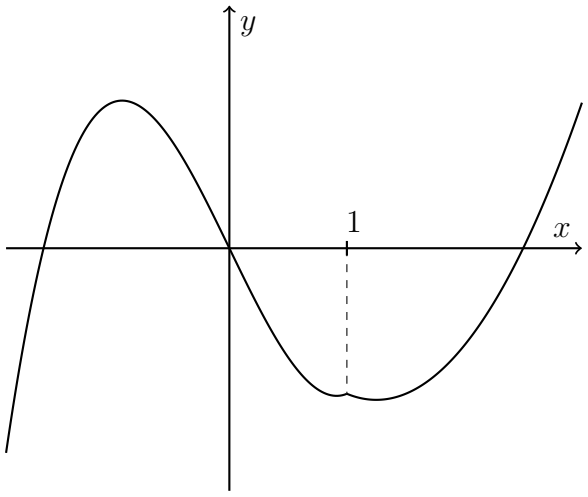
$x = 1$ is the turning point of $\text{Min}(x^2, x^3)$

Then $x = 1$ is also the turning point of y

When $x = 1, \quad y = -3 \implies (1, -3)$ is also a turning point

Hence the other three turning points are:

$$\boxed{\left(\frac{5}{4}, -\frac{25}{8}\right), \quad \left(\sqrt{\frac{5}{6}}, -\frac{10}{3}\sqrt{\frac{5}{6}}\right), \quad \left(-\sqrt{\frac{5}{6}}, \frac{10}{3}\sqrt{\frac{5}{6}}\right)}$$



This solution was submitted by **Camile**

Published and refined by **Jon**

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